CSCR2600 - Project-Based Learning (PBL) - 1

PROJECT TITLE

TOXIC – EXPENZO

Your AI-Powered Personal Finance Companion

B.TECH CSE (AI & ML) – YEAR: [Fill Year]

SEMESTER: [Fill Semester]

SESSION: 2026–2027

Submitted By:

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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1. Project Title

Toxic – Expenzo: Your AI-Powered Personal Finance Companion.

The title represents a proposed personal finance management application that helps users record transactions, monitor budgets, understand spending habits, and receive data-driven insights. The AI-powered features will be developed and tested as part of the project prototype.

2. Team / Group Formation

The project team should contain no more than three members. Fill in the official student details and assign roles before submission.

S. No.	Student Name	Roll Number	System ID	Role
1	[Enter name]	[Enter roll no.]	[Enter system ID]	Developer
2	[Enter name]	[Enter roll no.]	[Enter system ID]	Designer
3	[Enter name]	[Enter roll no.]	[Enter system ID]	Tester

Suggested roles: Developer – application and backend development; Designer – interface and user experience; Tester – test cases, validation, and defect logging.

3. Technologies to be Used

Software Platform

- **Front-end:** HTML5, CSS3, JavaScript; React may be used if a component-based interface is selected.
- **Back-end:** Python with Flask or FastAPI to process requests and apply business logic.
- **Database:** SQLite for a lightweight prototype, or MySQL if a multi-user database is required.
- **Data analysis and ML:** Pandas and NumPy for transaction analysis; Scikit-learn for suitable classification or prediction experiments.
- **Visualization:** Chart.js or Matplotlib for category-wise spending charts and monthly summaries.

Hardware Platform

- Computer/laptop with a modern multi-core processor, at least 4 GB RAM (8 GB recommended), and sufficient free disk space.
- Operating system: Windows, Linux, or macOS.
- Editor and browser: Visual Studio Code and a modern browser such as Chrome, Edge, or Firefox.

The exact technologies may be finalized according to available lab resources and implementation requirements.

4. Tools

Tool	Vendor/Provider	Purpose	Version
Visual Studio Code	Microsoft	Writing and debugging source code	Version used at development
Python	Python Software Foundation	Backend logic and data analysis	3.x
Flask / FastAPI	Pallets / FastAPI	Backend web API	Version used at development
SQLite / MySQL	SQLite / Oracle	Store user and transaction data	Version used at development
Git and GitHub	Git / GitHub	Version control and collaboration	Current version
Browser DevTools	Browser provider	Interface testing and debugging	Built-in

Version numbers should be updated to match the versions actually installed and used during development.

5. Problem Statement

Many people find it difficult to keep accurate records of daily income and expenses. Manual tracking can be inconsistent, and users may not notice which categories account for most of their spending. Without a clear view of their budget, they may exceed monthly limits or struggle to plan savings.

Toxic – Expenzo is proposed to address this problem through a centralized application where users can enter transactions, group expenses by category, set budgets, and view summaries in charts. Basic AI/ML-based analysis may identify patterns in the recorded data and provide general saving suggestions. The topic is selected because personal finance management is a practical everyday problem and offers an opportunity to apply software engineering, data analysis, and machine learning concepts.

The prototype will not require direct bank access. Users will enter transactions manually unless a secure, authorized integration is implemented separately.

6. Literature Survey

The following sources provide background on personal finance behaviour, budgeting, and the technical methods relevant to the proposed application. The literature review is preliminary and should be expanded if the faculty requires a wider research survey.

Author / Year	Work / Source	Relevant idea	Application to Expenzo
Thaler (1985)	Mental Accounting and Consumer Choice	People categorize money and expenses mentally, which can affect spending decisions.	Supports organizing transactions into categories and showing category-level summaries.
Soman (2001)	Effects of Payment Mechanism on Spending Behavior	Payment methods can influence spending behaviour and how people evaluate expenditure.	Motivates clear transaction records and spending feedback.
McKinney (2022)	Python for Data Analysis, 3rd ed.	Data analysis can help transform raw records into useful insights.	Supports summarizing transaction data and interpreting spending patterns.
Scikit-learn contributors	Scikit-learn User Guide	Describes machine-learning workflows, preprocessing, and model evaluation.	Provides guidance for optional categorization or forecasting experiments.

Research gap addressed: a simple student-level prototype can combine manual expense logging, budget monitoring, visual summaries, and introductory data-driven insights in one interface. Any ML feature must be evaluated using suitable data rather than assumed to be accurate.

7. Project Description

Expenzo is a proposed web-based personal finance companion. Its scope includes user transaction entry, income and expense totals, expense categories, budget limits, a dashboard, and graphical reports. A lightweight analysis component may provide rule-based insights or a machine-learning feature if adequate labelled or historical data is available.

The main system components are:

- **User interface:** forms for transactions and budgets, plus a dashboard.
- **Application backend:** validates entries, calculates totals, and applies budget rules.
- **Database:** stores users, transactions, categories, and budget targets.
- **Analysis module:** summarizes trends and, if implemented and tested, produces expense categorization or estimates.

High-Level Context Diagram

User ■ Expenzo Application ■ Transaction Database

Expenzo Application ■ Analysis / Insight Module → Dashboard & Budget Alerts

Inputs include transaction amount, date, type, category, description, and budget limits. Outputs include income/expense summaries, remaining balance, category-wise charts, budget alerts, and data-driven insights.

8. Project Modules: Design / Algorithm

Module 1 – User Access

Provide a login/registration interface if authentication is included. Validate inputs and keep each user's records separated.

Module 2 – Transaction Management

Allow the user to add, edit, delete, and view income and expense records. Validate that amounts and dates are reasonable.

Module 3 – Expense Categorization

Initially allow manual category selection. Optional ML categorization can be trained and evaluated using labelled transaction descriptions.

Module 4 – Budget Management

Let users define monthly and category-wise limits. Calculate budget utilization as (category spending ÷ category budget) × 100.

Module 5 – Dashboard and Reports

Calculate total income, total expenses, and remaining balance; display trends and category distributions using charts.

Module 6 – AI/ML Insights

Analyze historical transactions to identify repeated spending patterns. Forecasting should only be shown if enough historical data exists and model performance is evaluated.

Module 7 – Alerts and Recommendations

Trigger alerts when spending reaches a configurable threshold, and provide general suggestions based on observed patterns.

Basic calculation: Remaining Balance = Total Income – Total Expenses. Budget Utilization (%) = (Amount Spent ÷ Budget Limit) × 100. If a budget limit is zero or missing, utilization should not be calculated.

9. Implementation Methodology

- **Requirement analysis:** Confirm prototype features, user needs, constraints, and acceptance criteria.
- **Interface design:** Prepare wireframes for login (if used), dashboard, transaction form, budget page, and reports.
- **Database design:** Define entities such as User, Transaction, Category, and Budget, with appropriate relationships and validation rules.
- **Development:** Build the frontend, backend routes/API, database operations, calculations, and charts.
- **Analysis implementation:** Start with descriptive statistics and budget rules; add ML only if suitable data is available.
- **Testing:** Test valid and invalid inputs, calculations, transaction editing/deletion, budget thresholds, and chart outputs.
- **Defect log and maintenance:** Record each defect with an ID, description, steps to reproduce, severity, status, fix, and retest result. Resolve and retest defects before the demonstration.

Suggested testing cases include: adding a valid expense; rejecting a negative amount; calculating the remaining balance correctly; checking a budget threshold; editing and deleting a transaction; and confirming that charts reflect updated records.

Information flow: User enters a transaction → frontend validates basic input → backend validates and stores it → calculations and analysis run → dashboard updates with summaries and alerts.

10. Result & Conclusion

The expected result is a working prototype that records personal income and expenses, organizes transactions, monitors budgets, and presents financial summaries through a dashboard. It is also expected to demonstrate how data analysis can reveal spending patterns. The final results will depend on the features implemented and the testing performed.

The main contribution of Expenzo is the integration of expense tracking, budget monitoring, visual reporting, and introductory AI/ML insights in a single, accessible application. It is designed as a decision-support tool; its estimates and suggestions are not guaranteed financial advice.

11. Future Scope and Further Enhancement of the Project

- Develop an AI finance chatbot that answers questions about recorded transactions.
- Add receipt scanning using optical character recognition (OCR).
- Explore secure bank or payment-service integration only through supported, authorized interfaces.
- Improve expense forecasting with sufficient historical data and formal model evaluation.
- Add savings goals, recurring transactions, multi-currency support, and mobile access.
- Provide more detailed alerts and personalized budget recommendations while protecting user privacy.

12. Advantages of this Project

- Reduces the effort required to maintain expense records.
- Makes monthly income, expenses, and remaining balance easier to understand.
- Helps users compare spending against self-defined budgets.
- Presents category-wise spending through easy-to-read charts.
- Provides a practical platform for learning web development, database design, data analysis, and ML.
- May benefit students, households, and individuals who want to monitor everyday spending.

13. Outcome

Primary outcome: Project-to-Product prototype — a demonstrable web application for personal expense tracking and budget management.

Possible further outcomes, subject to the quality and completion of the work, include a hackathon demonstration or a research-oriented study on expense categorization and spending-pattern analysis. Publication, patents, or competition participation are possibilities only and are not guaranteed deliverables.

Demonstration checklist:

- Show transaction entry and transaction history.
- Show monthly income, expense, and balance calculations.
- Show a budget limit and corresponding alert.
- Show charts or reports based on sample transaction data.
- Demonstrate any AI/ML feature that has actually been implemented and tested.

14. References

- Thaler, R. H. (1985). Mental Accounting and Consumer Choice. Marketing Science, 4(3), 199–214. https://doi.org/10.1287/mksc.4.3.199
- Soman, D. (2001). Effects of Payment Mechanism on Spending Behavior: The Role of Rehearsal and Immediacy of Payments. Journal of Consumer Research, 27(4), 460–474. https://doi.org/10.1086/319621
- McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
- Python Software Foundation. Python Documentation. https://docs.python.org/3/
- The pandas development team. pandas Documentation. https://pandas.pydata.org/docs/
- Scikit-learn developers. Scikit-learn User Guide. https://scikit-learn.org/stable/
- Pallets Projects. Flask Documentation. https://flask.palletsprojects.com/
- React Team. React Documentation. https://react.dev/

Signature

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